Cricbuzz LiveStats Project

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1. INTRODUCTION

The Cricbuzz LiveStats project is an advanced cricket analytics dashboard built using Python, SQLite, and Streamlit. It enables users to explore match data, player statistics, and performance insights interactively

2. Objectives

1 Analyze cricket data using SQL queries

2 Develop an interactive dashboard using Streamlit

3 Visualize insights using charts and tables

4 Implement CRUD operations on database

5 Simulate real-world business analytics workflow

3. Tools & Technologies

- Python
- SQLite
- Streamlit
- Pandas
- VS Code

4. System Architecture

Data flows from SQLite database

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Processed using Python (Pandas)

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Visualized through Streamlit dashboard

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User interaction

Architecture Diagram

Database (SQLite)

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Python (Pandas)

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Streamlit Dashboard

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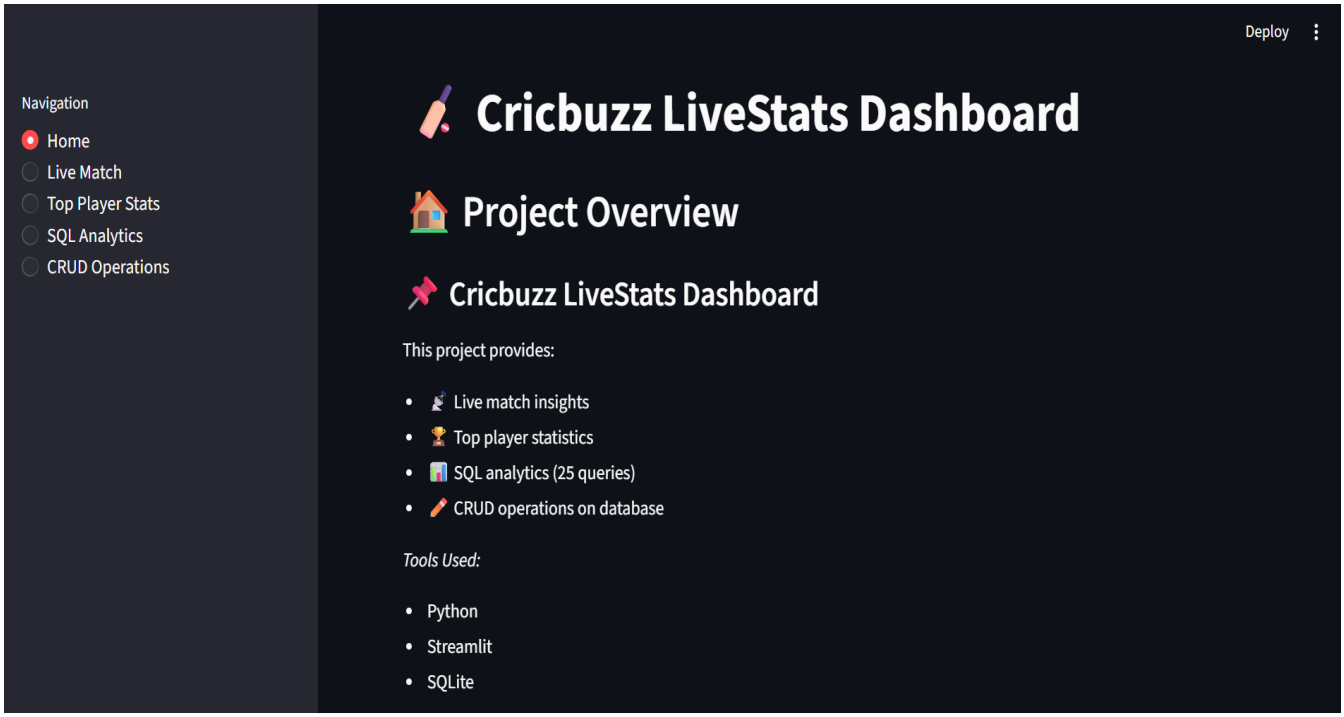
User Interaction

5. Dashboard Pages

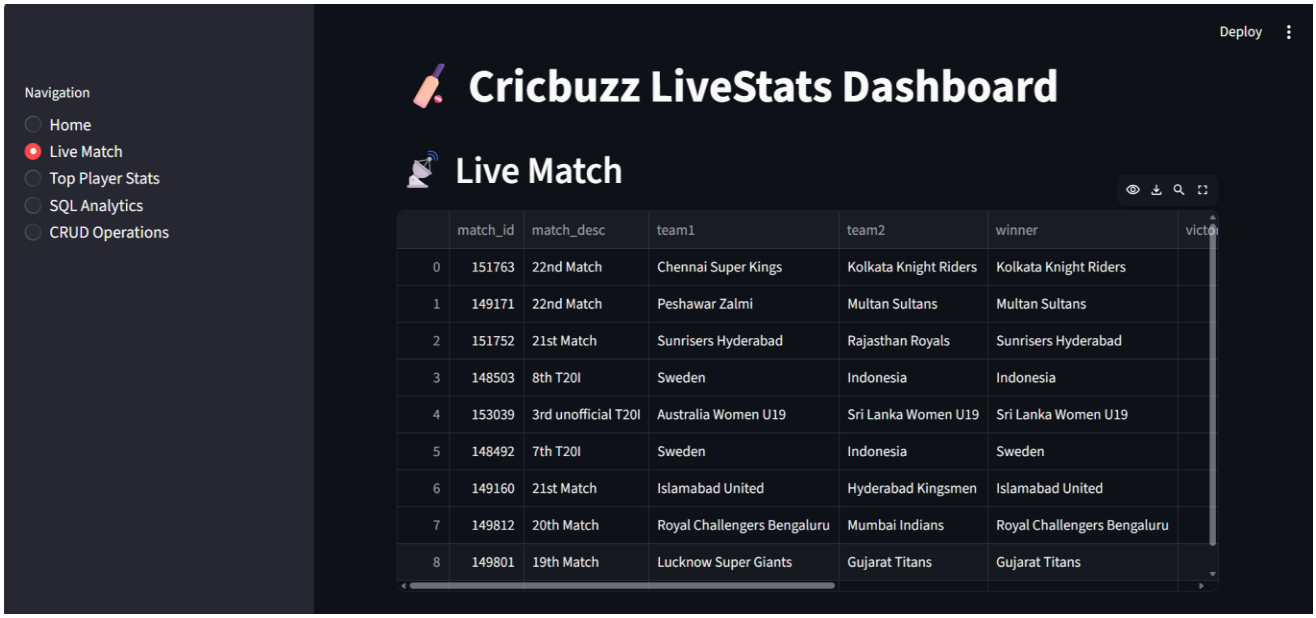
5.1 Home Page

Provides overview, navigation instructions, and project details.



5.2 Live Match Page

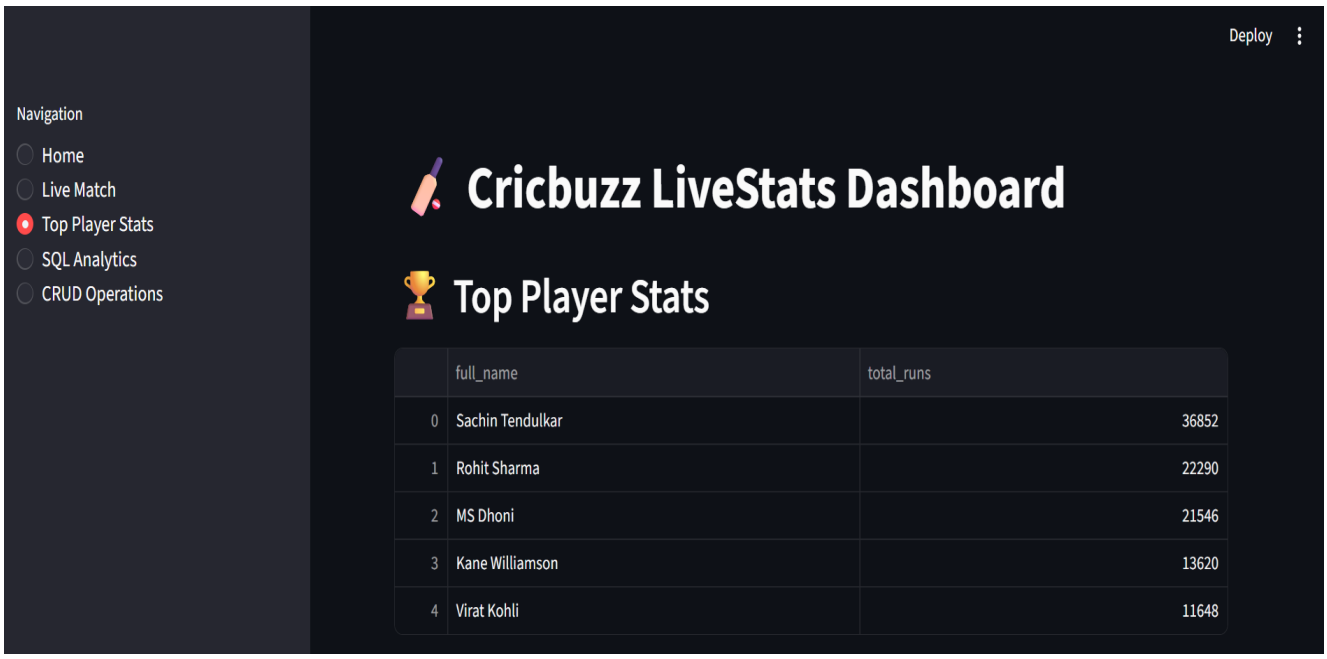
Displays match details including teams, venue, status, and results. Includes filters and summary metrics.



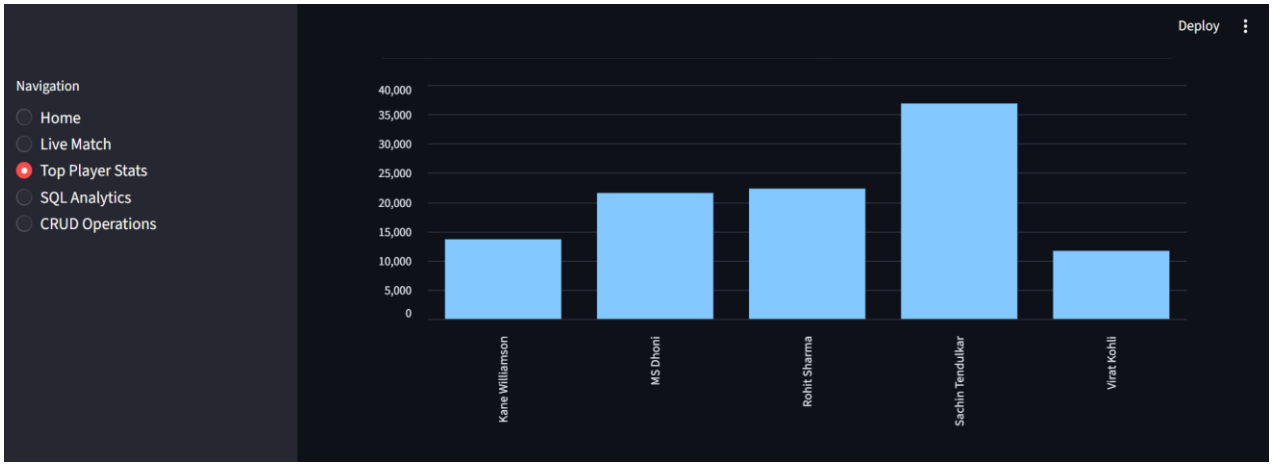
	match_id	match_desc	team1	team2	winner	victo
0	151763	22nd Match	Chennai Super Kings	Kolkata Knight Riders	Kolkata Knight Riders	
1	149171	22nd Match	Peshawar Zalmi	Multan Sultans	Multan Sultans	
2	151752	21st Match	Sunrisers Hyderabad	Rajasthan Royals	Sunrisers Hyderabad	
3	148503	8th T20I	Sweden	Indonesia	Indonesia	
4	153039	3rd unofficial T20I	Australia Women U19	Sri Lanka Women U19	Sri Lanka Women U19	
5	148492	7th T20I	Sweden	Indonesia	Sweden	
6	149160	21st Match	Islamabad United	Hyderabad Kingsmen	Islamabad United	
7	149812	20th Match	Royal Challengers Bengaluru	Mumbai Indians	Royal Challengers Bengaluru	
8	149801	19th Match	Lucknow Super Giants	Gujarat Titans	Gujarat Titans	

5.3 Top Player Stats Page

Shows top-performing players using charts and aggregated statistics.

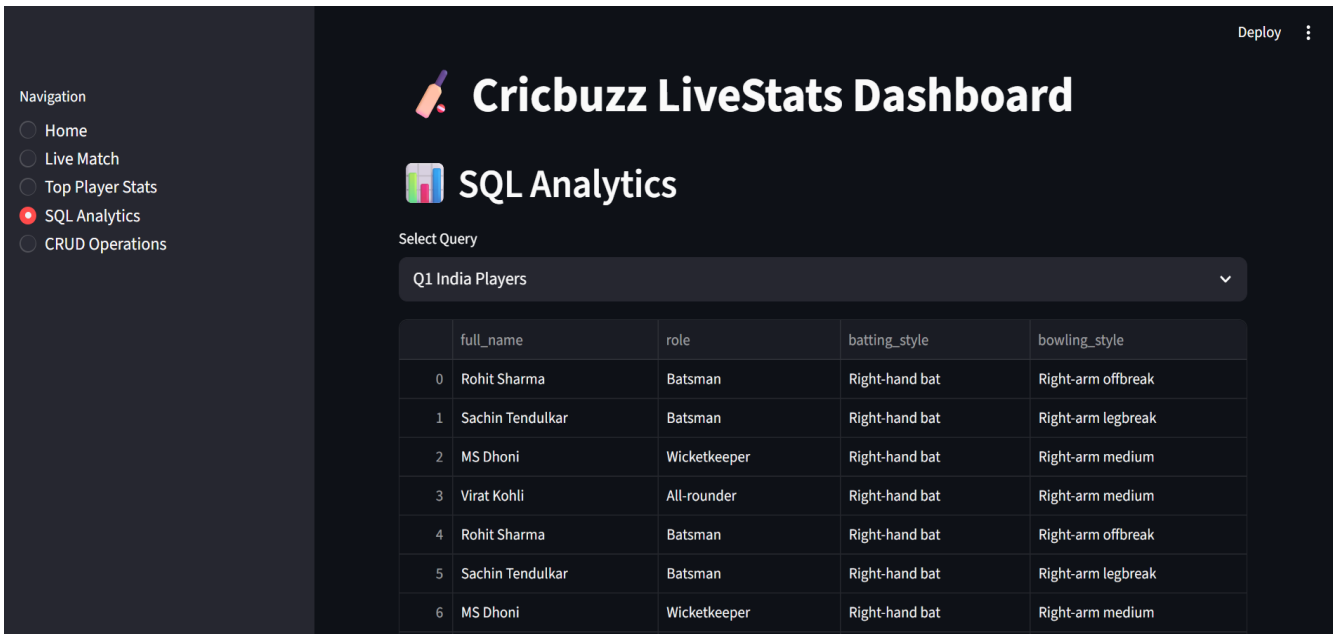


	full_name	total_runs
0	Sachin Tendulkar	36852
1	Rohit Sharma	22290
2	MS Dhoni	21546
3	Kane Williamson	13620
4	Virat Kohli	11648



5.4 SQL Analytics Page

Contains 25 advanced SQL queries providing insights into performance, trends, and match analytics.



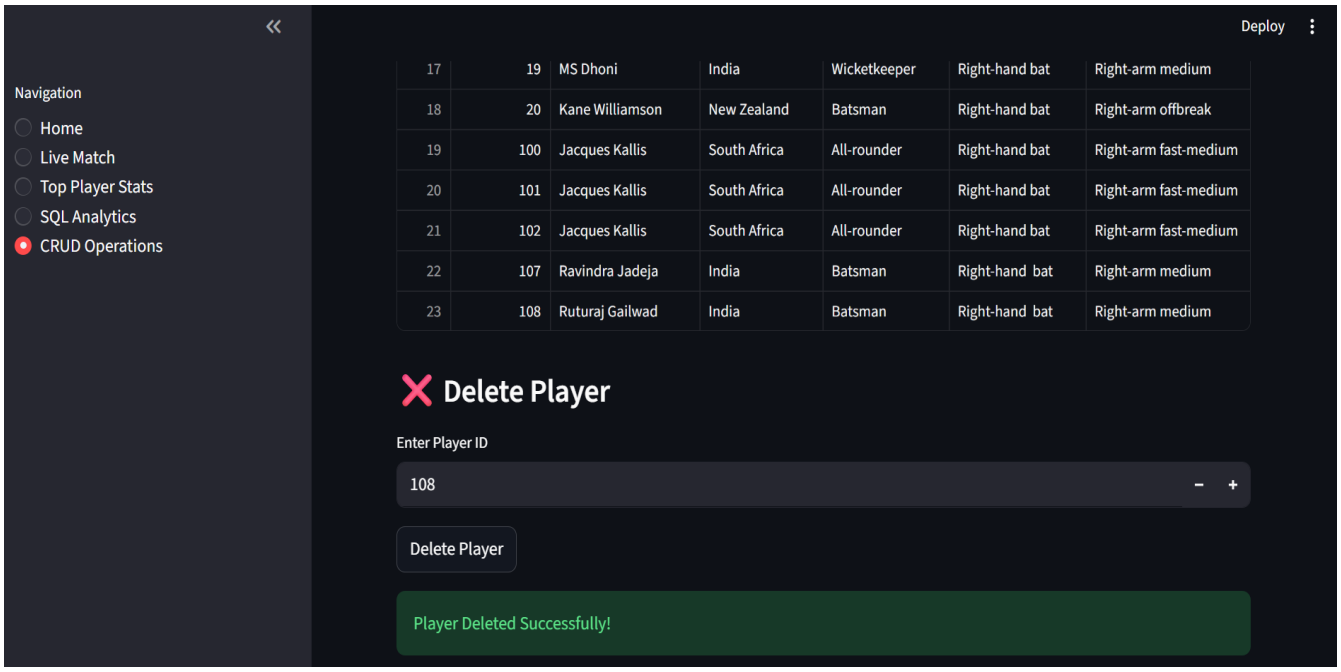
	full_name	role	batting_style	bowling_style
0	Rohit Sharma	Batsman	Right-hand bat	Right-arm offbreak
1	Sachin Tendulkar	Batsman	Right-hand bat	Right-arm legbreak
2	MS Dhoni	Wicketkeeper	Right-hand bat	Right-arm medium
3	Virat Kohli	All-rounder	Right-hand bat	Right-arm medium
4	Rohit Sharma	Batsman	Right-hand bat	Right-arm offbreak
5	Sachin Tendulkar	Batsman	Right-hand bat	Right-arm legbreak
6	MS Dhoni	Wicketkeeper	Right-hand bat	Right-arm medium

5.5 CRUD Operations Page

Implements Create, Read, Update, and Delete operations on player data.

ADD OPERATION

DELETE OPERATION



6. KEY SQL INSIGHTS

1 Top batsmen and bowlers performance

2 Toss impact analysis

3 Player consistency using standard deviation

4 Head-to-head team performance

5 Time-series performance trends

7. CHALLENGES AND SOLUTIONS

Challenges included handling null values, fixing SQL joins, resolving Streamlit errors, and correcting indentation issues. These were resolved using COALESCE, schema-based queries, debugging techniques, and proper code structuring.

8. CONCLUSION

This project demonstrates strong skills in SQL, Python, data analysis, and dashboard development, making it highly relevant for business analyst roles.